Experiment-2

Program:

```
import cv2
import matplotlib.pyplot as plt

def analyze_hist(img_path):
    img=cv2.imread(img_path)

    colors=('b', 'g', 'r')
    names=('Blue', 'Green', 'Red')

    plt.figure(figsize=(10,5))

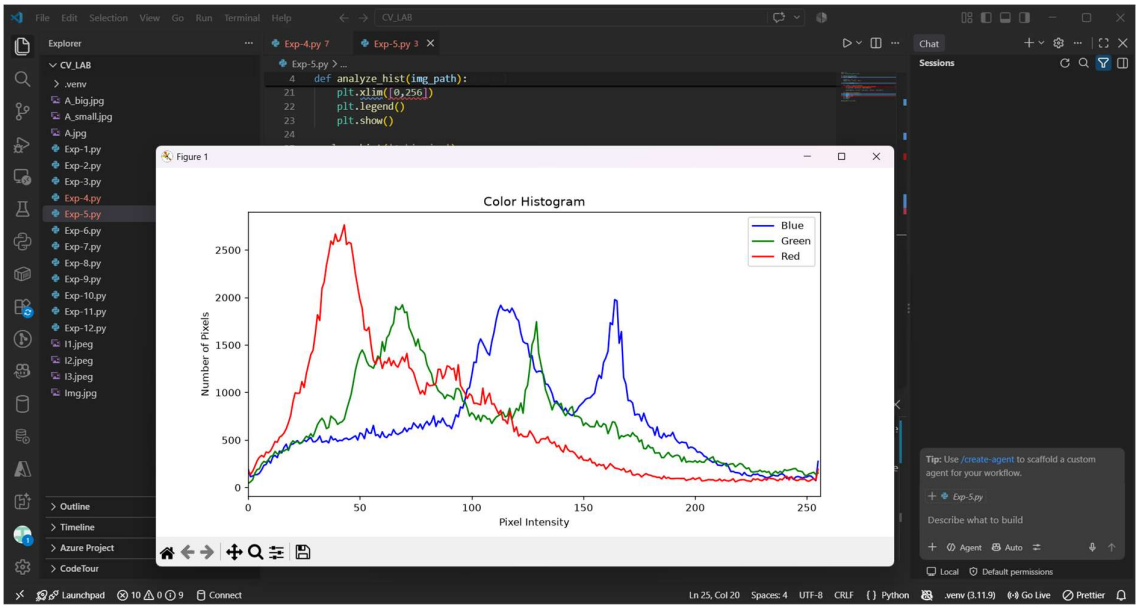
    for i,color in enumerate(colors):
        hist=cv2.calcHist([img], [i], None, [256], [0,256])
        plt.plot(hist, color=color, label=names[i])

        print(names[i], 'Minimum: ',hist.min(), 'Maximum: ',hist.max())

    plt.title("Color Histogram")
    plt.xlabel('Pixel Intensity')
    plt.ylabel('Number of Pixels')
    plt.xlim([0,256])
    plt.legend()
    plt.show()

analyze_hist('A_big.jpg')
```

Output:


Figure 1
Color Histogram
Blue
Green
Red
Number of Pixels
Pixel Intensity
0
500
1000
1500
2000
2500
0
50
100
150
200
250